EXPERIMENT NO.: - 9

Title: Software Engineering Tools and Technologies – Compare and Contrast Different Software Engineering Tools and Technologies

Program A: Login using Python

File1-login.py

```
STATIC_EMAIL = "test@gmail.com"
STATIC_PASSWORD = "123456"
MAX_ATTEMPTS = 3
def login(email, password):
    if email == STATIC_EMAIL and password == STATIC_PASSWORD:
            return True
    return False
print("========== LOGIN ==========")
attempts = 0
while attempts < MAX_ATTEMPTS:
      email = input("Enter Email    : ").strip()
      password = input("Enter Password : ").strip()
      if login(email, password):
       print("Login Successful")
       break
      else:
        attempts += 1
        print("Login Failed: Invalid email or password")
        print("Attempts left:", MAX_ATTEMPTS - attempts)
else:
        print("Too many failed attempts. Access blocked.")
```



Program B: Login using React.js + Node.js

The backend (Node.js + Express) holds the static email and password and checks them. The frontend (React.js) shows the login form and sends the data to the backend using a POST request.

File1- Server.js

```
const express = require("express");

const cors = require("cors");


const app = express();


// Middleware

app.use(cors());

app.use(express.json());


// Static login credentials

const STATIC_EMAIL = "test@gmail.com";

const STATIC_PASSWORD = "123456";


// Login API

app.post("/api/login", (req, res) => {

   const { email, password } = req.body || {};
```

```
  if (
    email === STATIC_EMAIL &&
    password === STATIC_PASSWORD
  ) {
    return res.status(200).json({
      success: true,
      message: "Login Successful"
    });
  }

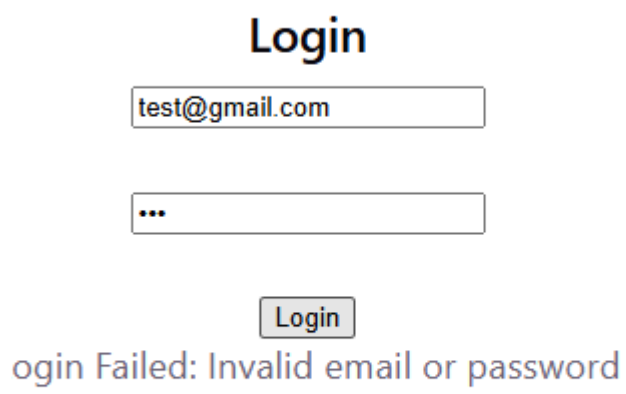
  return res.status(401).json({
    success: false,
    message: "Login Failed: Invalid email or password"
  });
});

// Start server
app.listen(5000, () => {
  console.log("Server running on http://localhost:5000");
});
```

File2: App.jsx

```
import { useState } from "react";
 export default function App() {   const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");   const [message, setMessage] = useState("");
   const [loggedIn, setLoggedIn] = useState(false);
     const handleLogin = async (e) => {     e.preventDefault();
       try {      const res = await fetch("http://localhost:5000/api/login", {
          method: "POST",        headers: { "Content-Type": "application/json" },
              body: JSON.stringify({ email, password }),      });
                  const data = await res.json();
```

```
                setMessage(data.message);      setLoggedIn(data.success);
                 } catch (err) {
                   setMessage("Server not reachable. Start the backend first.");
                    }  };
                     if (loggedIn) {    return (      <div style={{ textAlign: "center", marginTop: 80 }}>
                       <h2>Welcome, {email}</h2>       <p>{message}</p>       <button onClick={()
=> { setLoggedIn(false); setPassword(""); }}>         Logout       </button>      </div>    );  }
  return (    <form onSubmit={handleLogin}
style={{ maxWidth: 320, margin: "80px auto" }}>
 <h2>Login</h2>
    <input type="email" placeholder="Email" value={email}
       onChange={(e) => setEmail(e.target.value)} required />      <br /><br />      <input
type="password" placeholder="Password" value={password}
onChange={(e) => setPassword(e.target.value)} required />
  <br /><br />
 <button type="submit">Login</button>
  <p>{message}</p>    </form>   ); }
```



Welcome, test@gmail.com

Login Successful

Logout